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RESEARCH

Modified percutaneous vertebral body-intervertebral disc biopsy combined with genetic testing for the diagnosis of early-stage pyogenic spondylodiscitis: a retrospective cohort study

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Abstract

Objective This study aims to assess the diagnostic efficacy of a novel integrated approach that combines a modified percutaneous disc biopsy technique with 16 S rRNA gene sequencing to enhance pathogen identification in patients with early suspected pyogenic spondylodiscitis (PS).

Methods A retrospective cohort study was conducted, enrolling 66 patients who underwent percutaneous biopsy based on clinical and MRI findings indicative of early PS between January 2021 and January 2024. Patients were categorized into two groups: Group A ($n=31$), which received traditional transpedicular biopsy coupled with 16 S rRNA gene sequencing, and Group B ($n=35$), which received the modified transpedicular-vertebral-discal biopsy combined with genetic testing. A composite follow-up, including clinical symptoms, imaging, and laboratory findings at ≥6 months, served as the gold standard. Pathogen detection rates, complication rates, and pathogen profiles were compared between the groups.

Results Baseline characteristics were comparable between the groups. The pathogen detection rate in Group B (85.7%, 30/35) was significantly higher than that in Group A (51.6%, 16/31), with a statistically significant difference ($P<0.01$). No serious procedure-related complications were observed in either group. The predominant infection in both groups was monomicrobial bacterial infection, with *Staphylococcus aureus* identified as the most common pathogen.

Conclusion The modified percutaneous biopsy technique combined with 16 S rRNA gene sequencing significantly improves pathogen detection in early pyogenic spondylodiscitis without increasing the risk of complications. This integrated approach demonstrates substantial diagnostic value and shows promise as a first-line minimally invasive diagnostic method.

Keywords Pyogenic spondylodiscitis · Percutaneous biopsy · 16S rRNA gene sequencing · Pathogen detection · Minimally invasive diagnosis

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Abbreviations

PS	Pyogenic spondylodiscitis
BMI	Body Mass Index
CRP	C-Reactive Protein
ESR	Erythrocyte Sedimentation Rate
IQR	Interquartile Range

Introduction

Pyogenic spondylodiscitis (PS) represents a severe spinal infection [1], characterized by a relatively low incidence but rapid progression, which poses significant threats to patient health and quality of life [2]. "Pyogenic spondylodiscitis" describes infectious lesions affecting the intervertebral disc and adjacent vertebral bodies caused by microorganisms. PS can be classified based on the pathogen, with pyogenic bacteria such as *Staphylococcus aureus* and Streptococcus being the most common culprits, leading to acute suppurative inflammation. Other types include spinal tuberculosis caused by Mycobacterium tuberculosis and Brucellar spondylitis due to Brucella species, each exhibiting distinct epidemiological, clinical, radiological, and pathological characteristics (e.g., granulomatous inflammation). This study primarily focuses on enhancing the early etiological diagnosis of clinically atypical, early-stage spondylodiscitis suspected to result from common pyogenic bacteria. The clinical presentation of PS often manifests insidiously, with early symptoms such as back pain and fever being nonspecific and easily misinterpreted as other spinal conditions, resulting in delayed diagnoses [3]. Traditional diagnostic methods, which rely on clinical symptoms, imaging, and laboratory tests, exhibit notable limitations. While imaging can identify spinal lesions, it may reveal atypical features that are challenging to recognize in early-stage PS. Furthermore, imaging alone cannot determine the causative pathogen, potentially delaying treatment. Laboratory tests such as blood cultures frequently yield low positivity rates in early PS, as pathogens may not yet have disseminated, thus rendering these tests inadequate for early diagnosis [4]. These challenges contribute to diagnostic delays, increasing the risk of disability and exacerbating the socio-economic burden.

Recently, percutaneous vertebral biopsy has emerged as a widely accepted minimally invasive diagnostic tool for spinal diseases [5]. This technique facilitates direct procurement of lesion tissue under C-arm guidance, enabling both pathological diagnosis and pathogen detection [6]. However, conventional percutaneous biopsy techniques present certain drawbacks. The selection of puncture pathways and inherent technical complexities can elevate risks such as nerve injury or bleeding. Additionally, variable sample quality may compromise diagnostic accuracy. Traditional biopsy often retrieves only bone tissue samples, while evidence suggests that paravertebral soft tissues and intervertebral discs serve as optimal biopsy sites [7]. Moreover, relying solely on a single detection method frequently hampers the ability to achieve a rapid and accurate diagnosis of PS. The advent of genetic testing technologies has ushered in new prospects for diagnosing infectious diseases, allowing for rapid and precise pathogen nucleic acid detection with high sensitivity and specificity, thereby providing a robust foundation for clinical treatment [8, 9].

This study proposes a combined approach utilizing a modified percutaneous vertebral biopsy technique alongside genetic testing, with the goal of addressing the limitations of traditional methods and enhancing diagnostic accuracy and pathogen detection rates in early PS.

Mathods

This retrospective cohort study received approval from the hospital's Institutional Review Board (IRB) (Approval No.: 2023JX27-01). Given that the study involved retrospective data analysis and that all procedures were part of routine clinical care, the IRB waived the need for individual informed consent. The study adhered to the ethical principles outlined in the Declaration of Helsinki.

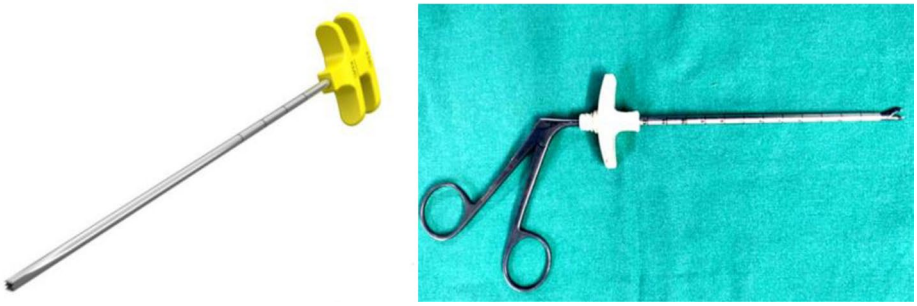
Medical records of patients admitted to the spine surgery department between January 2021 and January 2024, who underwent diagnostic evaluation for PS, were reviewed retrospectively. Inclusion criteria included: (1) Age≥18 years; (2) Clinical presentation of new-onset or worsening back pain, with or without fever, and MRI findings indicating abnormal signals in the intervertebral disc and adjacent vertebral bodies consistent with early PS; (3) Undergoing percutaneous vertebral biopsy with definitive pathological or microbiological results obtained; (4) Availability of ≥6 months of follow-up data (with clinical-imaging composite diagnosis as the gold standard). Exclusion criteria included: (1) Final diagnosis of spinal tuberculosis, brucellosis, tumor, or non-infectious lesions; (2) Receipt of effective antibiotic therapy within 2 weeks prior to biopsy; (3) Prior spinal instrumentation at the target level affecting the biopsy trajectory; (4) Missing or incomplete medical records.

Study groups and technical procedures

Based on the biopsy technique and detection strategy employed, the 66 patients included in the study were divided into two groups:

Group A (Traditional Biopsy + Genetic Testing, $n = 31$): All procedures were performed by the same spine surgeon

Fig. 1 Vertebral biopsy instrument



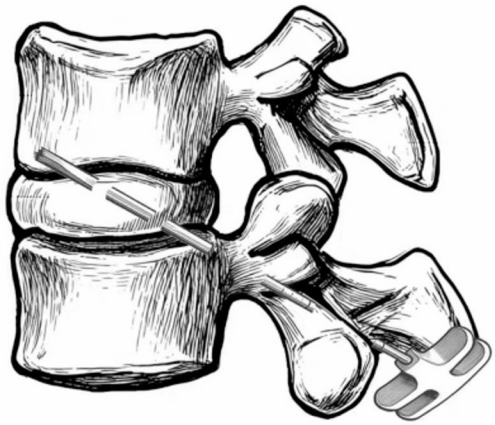


Fig. 2 Improved puncture biopsy schematic diagram. (**a**) Vertebral biopsy device, (**b**) Biopsy forceps

with over five years of experience under C-arm fluoroscopic guidance. Patients were positioned prone with the abdomen suspended. The C-arm was used to locate the pedicle of the target vertebral body on one side, and the surface entry point was typically at the 10 or 2 o'clock position relative to the pedicle shadow on the anteroposterior (AP) view. After routine disinfection, draping, and local anesthesia (approximately 5 ml of 1% lidocaine), the bone biopsy device (Fig. 1a) (Model: BC1101) was drilled into the osteolytic or bone-destructive area of the target vertebral body *via* the superolateral margin of the pedicle under continuous fluoroscopic monitoring. The accompanying biopsy instrument was then advanced into the bone lesion. Following a 360-degree rotation, the distal end of the biopsy device was closed to maintain a vacuum within the cannula. The device was subsequently withdrawn while rotating clockwise. The obtained tissue core was expelled using the inner stylet of the cannula. Biopsy forceps (Fig. 1b) were then utilized to supplementally retrieve additional tissue from the periphery of the lesion, aiming to acquire as much tissue as possible.

Group B (Modified Biopsy + Genetic Testing, $n = 35$): Patient positioning was identical to that in Group A. The C-arm was used to locate the pedicle of the target vertebral body. The skin entry point was planned based on the lateral fluoroscopic view and the location of the pathological area. On the AP view, this entry point was typically about 2 cm inferior to the pedicle shadow and 0.5–1 cm lateral to its outer margin. Routine disinfection, draping, and local anesthesia (approximately 5 ml of 1% lidocaine) were administered. Under continuous fluoroscopic monitoring, the biopsy needle (same as in Group A) was drilled into the bone lesion of the target vertebral body *via* the inferolateral margin of the pedicle. The accompanying biopsy device was then used to obtain sequential specimens following a "bone-disc-bone" approach (Fig. 2). Specifically, under fluoroscopic guidance, the biopsy device was first advanced into the bony lesion to obtain the initial specimen (Fig. 3a, b). It was then further advanced to penetrate the endplate using its serrated tip, entering the intervertebral disc space to collect disc tissue (Fig. 3c). Finally, depending on the individual lesion extent, the device could be advanced into the bony lesion of the adjacent superior vertebral body to obtain a third specimen (Fig. 3d). After a 360-degree rotation, the distal end was closed to maintain a vacuum within the cannula. The device was withdrawn while rotating clockwise, and the tissue cores were expelled using the inner stylet (Fig. 3e, f). Biopsy forceps were subsequently employed to collect additional bone and/or disc tissue, aiming to obtain as much material as possible (Fig. 3g, h).

Specimen processing and laboratory analysis

All tissue specimens obtained from both groups were immediately sent to the laboratory under sterile conditions for parallel processing. Each specimen was divided into three parts for: (1) Microbial Culture: This was performed according to standard operating procedures for both aerobic and anaerobic cultures for a minimum of 5 days; (2) Pathological Examination: Tissues were fixed in formalin, embedded in paraffin, sectioned, and stained with Hematoxylin and Eosin (H&E) and Gram stain. Two senior pathologists, blinded to the clinical groups, independently assessed the presence of acute inflammatory cell infiltration and pathogen morphology; (3) Genetic Testing: Pathogen identification was conducted using 16 S rRNA gene amplicon sequencing. DNA

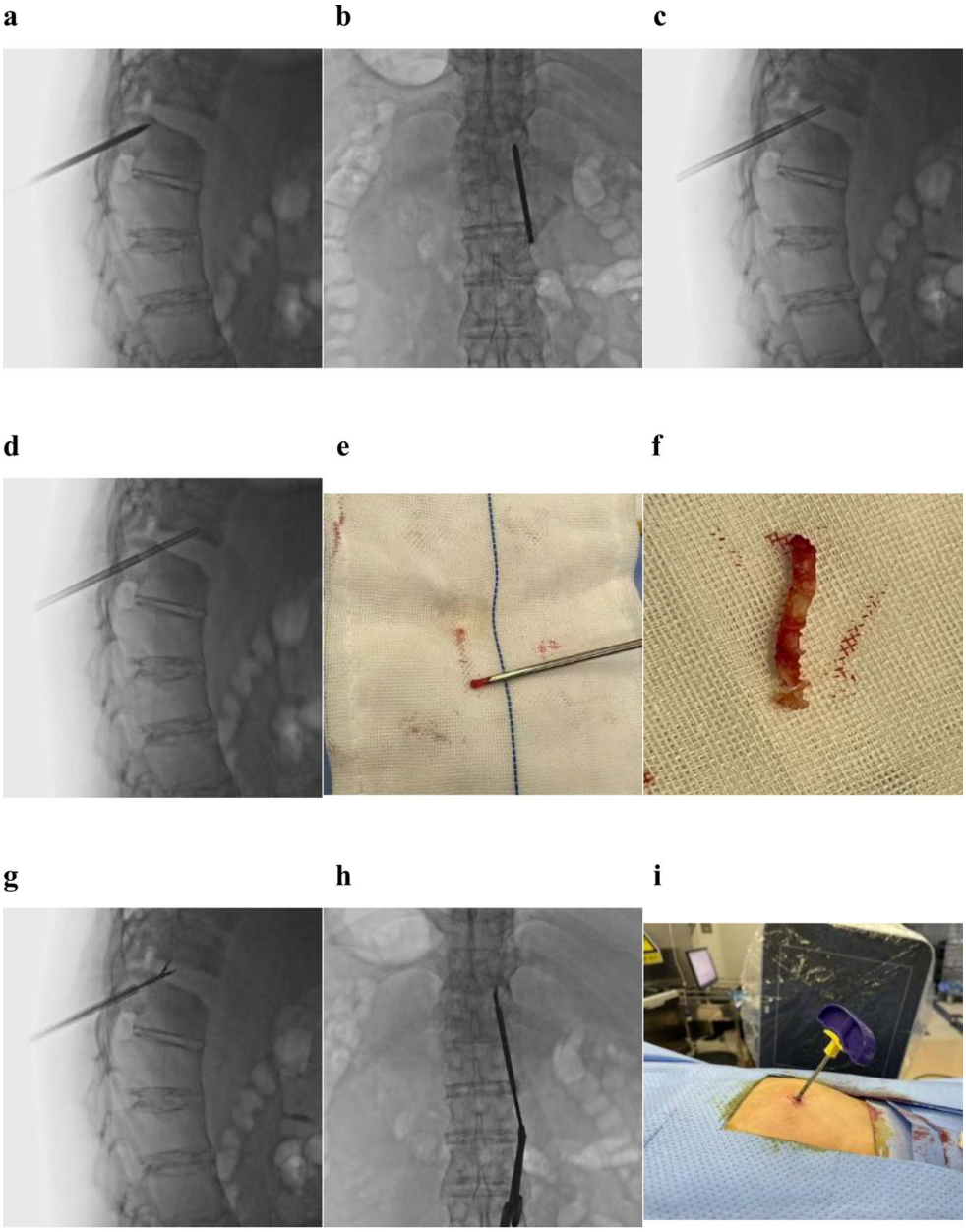


Fig. 3 Schematic diagram of the modified percutaneous vertebral body-intervertebral disc biopsy. (**a**, **b**) Anteroposterior and lateral fluoroscopic views showing the biopsy needle positioned within the infected vertebral body; (**c**, **d**) Lateral views showing the needle advanced into the adjacent intervertebral disc and vertebral body; (**e**, **f**) Macroscopic appearance of the obtained tissue specimens from the vertebral body and intervertebral disc, respectively. (**g**, **h**) Use of biopsy forceps for supplementary sampling; (**i**) Schematic illustration of the patient's prone position during the procedure

extracted from the samples underwent PCR amplification, library preparation, and high-throughput sequencing on an Illumina platform. The obtained sequences were compared against databases to identify the pathogen species.

Outcome measures and gold standard definition

A composite follow-up result at ≥6 months served as the final diagnostic gold standard. This composite result included: (1) Symptom relief after antibiotic therapy, normalization of inflammatory markers, and reduction or resolution of MRI lesions; (2) Intraoperative pathological/microbiological confirmation; or (3) If the follow-up did not meet the above criteria, the case was considered negative.

Statistical analysis was performed using SPSS 25.0 software. Normally distributed continuous data were expressed as mean ± standard deviation ($\bar{x} \pm s$) and compared between groups using the independent samples t-test. Non-normally distributed continuous data were presented as median (interquartile range) [M (IQR)] and compared using the Mann-Whitney U test. Categorical data were presented as number (percentage) [n (%)] and compared using the Chi-square test or Fisher's exact test. A P-value < 0.05 was considered statistically significant.

Table 1 Baseline Demographic and Clinical Characteristics of the Study Cohort

Characteristic	Modified Group ($n=35$)	Traditional Group ($n=31$)	P value
Age, years, mean ± SD (range)	67.7 ± 10.4	68.3 ± 7.0	0.892
Sex, n (%)			
Male	22 (62.9)	16 (51.6)	0.621
Female	13 (37.1)	15 (48.4)	
BMI, kg/m^2, mean ± SD	23.6 ± 4.4	23.0 ± 3.7	0.314
Pre-op CRP, mg/L, Median [IQR]	15.51 [6.63, 46.05]	15.51 [1.92, 46.4]	0.457
Pre-op ESR, mm/hr, Median [IQR]	46 [26, 99]	67 [19, 86]	0.679
Involved level, n (%)			
Cervical (C1-C7)	0 (0)	0 (0)	---
Thoracic (T1-T12)	11 (31.4)	7 (22.6)	-
Lumbar (L1-L5)	19 (54.3)	21 (67.7)	-
Sacral (S1)	5 (14.3)	3 (9.7)	-
Diabetes, n (%)	11 (31.4)	6 (19.4)	0.543

BMI, Body Mass Index; CRP, C-Reactive Protein; ESR, Erythrocyte Sedimentation Rate; IQR, Interquartile Range

Table 2 Pathogen Spectrum among Microbiologically Confirmed Cases

Category	Modified Group ($n=32$ positive), n (%)	Traditional Group ($n=17$ positive), n (%)
Bacteria only	15 (46.9)	11 (64.7)
Fungi only	1 (3.1)	1 (5.9)
Virus only	2 (6.2)	1 (5.9)
Polymicrobial	14 (43.8)	4 (23.5)
Most common isolate	*Staphylococcus aureus*	*Staphylococcus aureus*

Table 3 Comparison of Positive Diagnostic Rates between Groups

Group	Sample Size (n)	Positive Cases (n)	Positive Rate (%)	P value
A: Traditional	31	16	51.6%	$P<0.01$
B: Modified	35	30	85.7%	

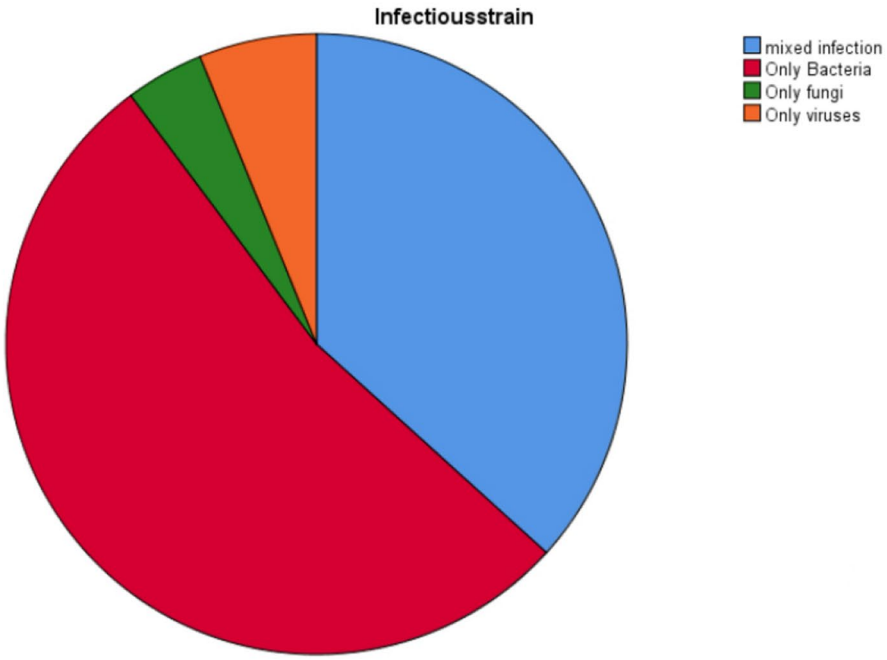
Results

A total of 66 patients met the inclusion criteria for analysis. The traditional biopsy + genetic testing group (Group A) comprised 31 patients (48.4% female), while the modified biopsy + genetic testing group (Group B) included 35 patients (42.9% female). No significant differences were observed between the two groups concerning age, sex, BMI, preoperative CRP, ESR, distribution of involved vertebral levels, or prevalence of diabetes ($P>0.05$), indicating comparable baseline characteristics (Table 1). The pathogen spectra were also similar between the groups (Table 2). The distribution of pathogen types was consistent in both groups, with monomicrobial bacterial infection being the most common cause of PS, and *S. aureus* identified as the predominant pathogen (Table 2).

For the final analysis, cases in which only viral sequences were detected by 16 S rRNA gene sequencing, without co-detection of bacteria or fungi, were classified as microbiologically negative. This classification was based on the following considerations: (1) The core objective of this study is to enhance the etiological diagnosis of PS, which is primarily a bacterial spinal infection; (2) Current clinical and pathological consensus suggests that viruses alone are highly unlikely to be the cause of classic PS, with their detection more likely reflecting reactivation of latent viruses or asymptomatic colonization rather than the etiology of the current infectious lesion.

The positive pathogen detection rate in the modified biopsy + genetic testing group (Group B) was 85.7% (30/35), significantly higher than the rate of 51.6% (16/31) in the traditional biopsy + genetic testing group (Group A), showing a statistically significant difference ($P<0.01$) (Table 3). Among the 49 specimens with detected microbial or viral nucleic acids, *S. aureus* was the most frequently identified bacterial pathogen, followed by various other bacterial genera. Human herpesviruses (HHV) and torque teno viruses (TTV) were also identified in a subset of samples; however, these viral sequences were predominantly detected in the presence of concomitant bacterial DNA. In cases where only viral sequences were present without bacterial or fungal co-detection, such cases were classified as microbiologically negative for diagnosing PS, as current evidence does not support viruses as primary causative agents of this condition (Fig. 4). No serious complications, such as nerve injury, paravertebral hematoma, or trajectory infection, occurred in either group. All patients completed a follow-up period of ≥6 months. Ultimately, 62 patients (93.9%) met

a


Infectiousstrain
mixed infection
Only Bacteria
Only fungi
Only viruses

b

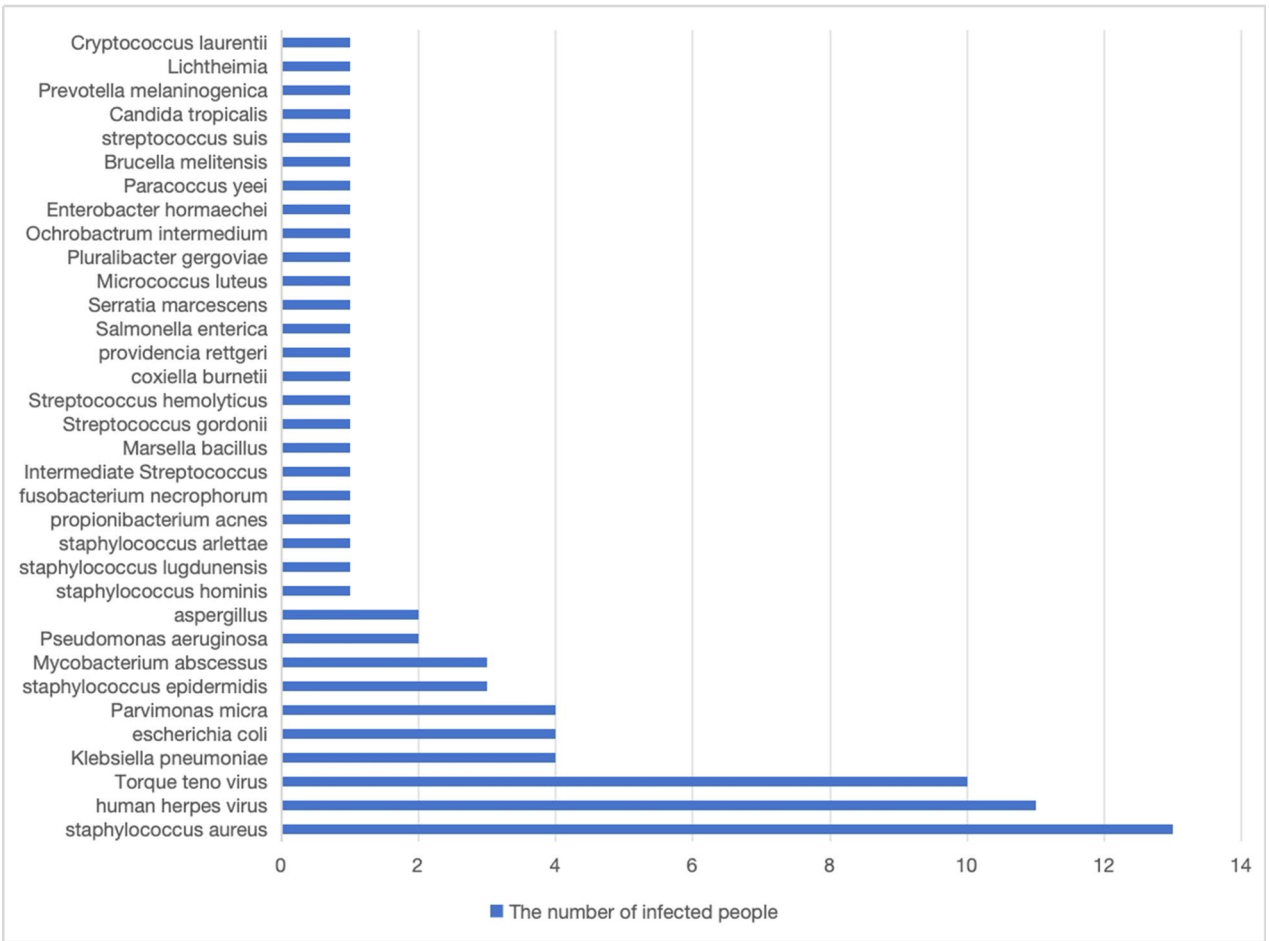

Cryptococcus laurentii
Lichtheimia
Prevotella melaninogenica
Candida tropicalis
streptococcus suis
Brucella melitensis
Paracoccus yeei
Enterobacter hormaechei
Ochrobactrum intermedium
Pluralibacter gergoviae
Micrococcus luteus
Serratia marcescens
Salmonella enterica
providencia rettgeri
coxiella burnetii
Streptococcus hemolyticus
Streptococcus gordonii
Marsella bacillus
Intermediate Streptococcus
fusobacterium necrophorum
propionibacterium acnes
staphylococcus arlettae
staphylococcus lugdunensis
staphylococcus hominis
aspergillus
Pseudomonas aeruginosa
Mycobacterium abscessus
staphylococcus epidermidis
Parvimonas micra
escherichia coli
Klebsiella pneumoniae
Torque teno virus
human herpes virus
staphylococcus aureus
0
2
4
6
8
10
12
14
The number of infected people

Fig. 4 Pathogen Profile in Patients with Pyogenic Spondylodiscitis. (**a**) Distribution of pathogens in all positive cases; (**b**) Distribution of infecting bacterial genera in all positive cases

the clinical-imaging composite remission criteria following antibiotic or combined surgical treatment. Four patients (2 in Group A, 2 in Group B) underwent secondary surgical instrumentation due to multi-level destruction and recovered well postoperatively. Additionally, no deaths were reported. In summary, the modified posterolateral approach combined with genetic testing significantly enhanced the pathogen detection rate for early PS without increasing the risk of complications, demonstrating reliable clinical application value.

Discussion

The results of this study demonstrate that, compared to the traditional transpedicular approach, the modified posterolateral approach, which facilitates continuous "bone-disc-bone" sampling combined with 16 S rRNA gene sequencing, significantly enhances the pathogen detection rate for early PS (85.7% vs. 51.6%) without increasing procedure-related risks. This finding introduces a powerful new strategy for the precise diagnosis of early PS.

Early diagnosis of PS remains a major clinical challenge. Although traditional microbial culture is considered the "gold standard," its positivity rate is often hindered by factors such as low pathogen load, prior antibiotic use, and the presence of difficult-to-culture microorganisms (e.g., anaerobes, fastidious organisms). Reported positivity rates typically range from 30% to 70% [10]. The 51.6% detection rate in the traditional biopsy group of this study aligns with this range, highlighting the limitations of traditional methods. In recent years, molecular diagnostic techniques, particularly 16 S rRNA gene amplicon sequencing, have demonstrated considerable promise in bone and joint infections due to their high sensitivity, ability to detect pathogens without pre-culture, and capacity to identify rare microorganisms. This study innovatively combined this advanced diagnostic technology with an enhanced biopsy sampling method, resulting in a significant improvement in diagnostic performance [11–13].

The significantly improved detection rate in the modified biopsy group can be attributed to the synergistic effect of two key factors. First, the modified biopsy trajectory played a pivotal role in successfully obtaining representative specimens. The pathological process of PS typically begins in the intervertebral disc before spreading to the adjacent vertebral endplates and paravertebral soft tissues. The traditional transpedicular approach primarily samples intravertebral tissue, which, in the early stages of the disease, may not contain sufficient pus or necrotic tissue rich in pathogens, resulting in inadequate sampling [14, 15]. In contrast, the modified percutaneous vertebral body-intervertebral disc biopsy technique used in this study allows for continuous "bone-disc-bone" sampling, simultaneously collecting tissue from the inflammatory core of the disc and the lesioned tissue in the endplate region of the adjacent vertebral body. This significantly increases the likelihood of obtaining viable pathogen specimens, thereby enhancing detection sensitivity at the source. Second, genetic sequencing technology ensures a high detection rate. 16 S rRNA sequencing does not rely on the viability of the pathogen; it only requires the presence of trace amounts of pathogen DNA in the sample [16]. This overcomes the challenges posed by negative cultures following antibiotic use and allows for the detection of difficult-to-culture pathogens [17].

The high sensitivity of 16 S rRNA gene sequencing, while enhancing bacterial detection, also introduces challenges in result interpretation. In this study, viral sequences—specifically HHV and TTV—were identified in a subset of samples. However, the detection of viral DNA does not equate to a diagnosis of viral spondylodiscitis. Primary viral spondylodiscitis is exceedingly rare and not well documented in the literature. Instead, the presence of these viral sequences most likely reflects one of the following scenarios: (1) reactivation of latent viruses in the setting of local inflammation or altered immune status; (2) asymptomatic viral colonization or bystander presence in infected tissue [18]; or (3) background nucleic acid contamination, which can occur with highly sensitive molecular methods. Therefore, for the primary objective of this study—improving the etiological diagnosis of bacterial PS—cases with viral sequences alone, without concurrent bacterial or fungal detection, were conservatively classified as microbiologically negative. This conservative approach avoids overdiagnosis and aligns with current clinical and pathological consensus that viruses are not established causative agents of primary PS. Future studies employing metagenomic sequencing and host transcriptional profiling may help clarify the biological significance of viral DNA detection in this context [19].

The absence of serious complications in both groups highlights the safety of the modified biopsy technique. When performed by an experienced surgeon, this approach did not increase the risk of neurovascular injury or infection. This favorable safety profile can be attributed to the clear fluoroscopic guidance and precise understanding of anatomical structures.

This study has several limitations. First, as a single-center retrospective study, despite strict inclusion and exclusion criteria, selection bias may still be present. Furthermore, due to geographical factors and the specific inclusion criteria focusing on early pyogenic presentations, pathogens

such as Brucella and Mycobacterium tuberculosis—important causes of spondylodiscitis in other regions or clinical contexts—were not represented in this cohort. Second, the relatively limited sample size may reduce the statistical power for analyzing certain rare pathogen subgroups. Third, the high sensitivity of gene sequencing requires caution against potential false-positive results from environmental contamination or background nucleic acids. Results must be interpreted in conjunction with clinical, imaging, and other laboratory findings. Fourth, while this study primarily focused on diagnostic efficacy, the impact of this strategy on long-term patient outcomes and its cost-effectiveness must be validated through prospective, multicenter, long-term follow-up studies.

In conclusion, modified percutaneous biopsy combined with 16 S rRNA gene sequencing is an efficient and safe diagnostic strategy. By optimizing the sampling site and leveraging advanced molecular techniques, it effectively addresses the clinical challenge of low pathogen detection rates in early PS. This systematic approach, focused on enhancing pathogen identification, provides a practical pathway for the standardized diagnosis of early suspected PS. Future research should further assess the applicability of this strategy in multicenter settings and its impact on long-term patient outcomes.

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Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by J.Y.G. X.M.Y.L.C.P. prepared figures and table . All authors commented on previous versions of the manuscript and read and approved the final version.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval This research was approved by the Medical Ethics Review Committee of Jiaxing Second Hospital (2023JX27-01).

Consent to participate Not applicable.

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